

Complete Slide Plans Index

Modules 1 - 4 (Sessions 1 to 59)

Low-Voltage Systems, Security & Smart Home Automation Curriculum

Compiled: July 2026

Module 1 Slide Plans Outline

Session 1: Electrical Fundamentals AC/DC

Focus: Voltage, current, resistance, Ohm's law, and SPDT switch circuits.

- **(1-1) Title Slide:** Introductory splash screen for the Electrical Fundamentals AC/DC session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(1-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(1-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(1-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(1-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(1-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(1-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(1-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(1-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(1-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 2: Earthing & Surge Protection

Focus: GI pipe pit layers, SPDs, and Earth test lamp testing.

- **(2-1) Title Slide:** Introductory splash screen for the Earthing & Surge Protection session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(2-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.

** Diagram: Concept block diagram mapping input-process-output stages.*

- **(2-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.

** Diagram: Detailed schematic showing pin configurations and physical links.*

- **(2-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.

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- **(2-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.

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** Diagram: Electrical circuit isolating ground terminals and surges.*

- **(2-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.

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- **(2-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.

** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*

- **(2-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.

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Session 3: Multimeter Diagnostics

Focus: DMM probe ports, continuity trace, and short circuit fault-finding.

- **(3-1) Title Slide:** Introductory splash screen for the Multimeter Diagnostics session.

** Diagram: Center-aligned splash screen with low-voltage course tag branding.*

- **(3-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.

** Diagram: Concept block diagram mapping input-process-output stages.*

- **(3-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.

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Session 4: Electrical Safety & Breakers

Focus: Shock thresholds, LOTO rules, MCB/RCCB leakage trip tests.

- **(4-1) Title Slide:** Introductory splash screen for the Electrical Safety & Breakers session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(4-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
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Session 5: Power Supply & SMPS

Focus: Linear vs SMPS, battery sizing calculations, pot calibrations.

- **(5-1) Title Slide:** Introductory splash screen for the Power Supply & SMPS session.

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- **(5-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
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Session 6: Reading Drawings

Focus: Symbols and grid coordination, relays, dry vs wet contacts.

- **(6-1) Title Slide:** Introductory splash screen for the Reading Drawings session.
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Session 7: Cable Selection & Conduits

Focus: AWG vs metric, voltage drop, 40% conduit fill pull tests.

- **(7-1) Title Slide:** Introductory splash screen for the Cable Selection & Conduits session.
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Session 8: IP Addressing & Subnetting

Focus: IPv4 subnet CIDR masks, default gateways, static IP setup.

- **(8-1) Title Slide:** Introductory splash screen for the IP Addressing & Subnetting session.
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Session 9: Router & Switch Configuration

Focus: WAN/LAN setups, NAT port forwarding, Wi-Fi channels.

- **(9-1) Title Slide:** Introductory splash screen for the Router & Switch Configuration session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
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Session 10: VLAN Basics

Focus: Broadcast segment separation, 802.1Q tags, access trunk ports.

- **(10-1) Title Slide:** Introductory splash screen for the VLAN Basics session.
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Session 11: PoE Switches

Focus: 802.3af/at/bt wattage levels, active/passive PoE loops.

- **(11-1) Title Slide:** Introductory splash screen for the PoE Switches session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(11-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
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Session 12: Fiber Optic Basics

Focus: Single-mode vs Multi-mode, SFP transceiver speeds, media converters.

- **(12-1) Title Slide:** Introductory splash screen for the Fiber Optic Basics session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(12-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
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- **(12-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
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Session 13: Network Troubleshooting

Focus: RJ45 crimping, wiremap tester, loop storms, IP conflicts.

- **(13-1) Title Slide:** Introductory splash screen for the Network Troubleshooting session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(13-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
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Module 2 Slide Plans Outline

Session 14: Analog & IP CCTV

Focus: Coaxial transmission vs Cat6 UTP, HD-TVI protocol, video baluns.

- **(14-1) Title Slide:** Introductory splash screen for the Analog & IP CCTV session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(14-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(14-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
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Session 15: Camera Selection & Design

Focus: Dome, Bullet, and Turret shapes, True WDR, DORI target levels.

- **(15-1) Title Slide:** Introductory splash screen for the Camera Selection & Design session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(15-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.

** Diagram: Concept block diagram mapping input-process-output stages.*

- **(15-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.

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Session 16: Lens & Field of View

Focus: Focal lengths wide vs telephoto, optical zoom, FOV calculator.

- **(16-1) Title Slide:** Introductory splash screen for the Lens & Field of View session.

** Diagram: Center-aligned splash screen with low-voltage course tag branding.*

- **(16-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.

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- **(16-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.

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Session 17: DVR, NVR & Hybrids

Focus: SATA HDD installations, PoE switch chassis, recording profiles.

- **(17-1) Title Slide:** Introductory splash screen for the DVR, NVR & Hybrids session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(17-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
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Session 18: RAID & Storage Design

Focus: RAID 0/1/5 redundancy, hot-swapping drives, storage retention.

- **(18-1) Title Slide:** Introductory splash screen for the RAID & Storage Design session.

** Diagram: Center-aligned splash screen with low-voltage course tag branding.*

- **(18-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
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Session 19: ONVIF Standards

Focus: Profile S, G, T cross-brand device integrations discovery.

- **(19-1) Title Slide:** Introductory splash screen for the ONVIF Standards session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(19-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(19-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(19-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(19-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(19-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*

- **(19-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(19-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(19-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(19-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 20: Video Compression

Focus: H.264 vs H.265+ codecs, CBR/VBR bandwidth monitoring.

- **(20-1) Title Slide:** Introductory splash screen for the Video Compression session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(20-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(20-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(20-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(20-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(20-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(20-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(20-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(20-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(20-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 21: AI Analytics

Focus: Line crossing rules, intrusion zone alert settings, human filters.

- **(21-1) Title Slide:** Introductory splash screen for the AI Analytics session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(21-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(21-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(21-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(21-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(21-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(21-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(21-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(21-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(21-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 22: ANPR Cameras

Focus: Shutter speeds, camera mounting angles, plate whitelists.

- **(22-1) Title Slide:** Introductory splash screen for the ANPR Cameras session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(22-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(22-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(22-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*

- **(22-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
* *Diagram: Configuration database schema or address map block.*
- **(22-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
* *Diagram: Electrical circuit isolating ground terminals and surges.*
- **(22-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
* *Diagram: Mounting dimension wall map indicating clearance codes.*
- **(22-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
* *Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(22-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
* *Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(22-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
* *Diagram: Lab trainer schematic showing green verification check.*

Session 23: Facial Recognition

Focus: Face pixel enrollment, database matching thresholds, alerts.

- **(23-1) Title Slide:** Introductory splash screen for the Facial Recognition session.
* *Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(23-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
* *Diagram: Concept block diagram mapping input-process-output stages.*
- **(23-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
* *Diagram: Detailed schematic showing pin configurations and physical links.*
- **(23-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
* *Diagram: Power loop routing circuit showing active voltage paths.*
- **(23-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
* *Diagram: Configuration database schema or address map block.*
- **(23-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
* *Diagram: Electrical circuit isolating ground terminals and surges.*
- **(23-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
* *Diagram: Mounting dimension wall map indicating clearance codes.*
- **(23-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
* *Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(23-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
* *Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(23-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.

** Diagram: Lab trainer schematic showing green verification check.*

Session 24: ColorVu & Thermal Cameras

Focus: Night supplemental lighting, thermal heat source point alarms.

- **(24-1) Title Slide:** Introductory splash screen for the ColorVu & Thermal Cameras session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(24-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(24-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(24-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(24-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(24-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(24-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(24-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(24-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(24-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 25: PTZ Cameras

Focus: Servo motor slip rings, preset coordinates, continuous scans.

- **(25-1) Title Slide:** Introductory splash screen for the PTZ Cameras session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(25-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(25-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*

- **(25-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
* *Diagram: Power loop routing circuit showing active voltage paths.*
- **(25-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
* *Diagram: Configuration database schema or address map block.*
- **(25-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
* *Diagram: Electrical circuit isolating ground terminals and surges.*
- **(25-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
* *Diagram: Mounting dimension wall map indicating clearance codes.*
- **(25-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
* *Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(25-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
* *Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(25-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
* *Diagram: Lab trainer schematic showing green verification check.*

Session 26: Remote Viewing & Cloud

Focus: P2P cloud engines, mobile links QR code, DDNS maps.

- **(26-1) Title Slide:** Introductory splash screen for the Remote Viewing & Cloud session.
* *Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(26-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
* *Diagram: Concept block diagram mapping input-process-output stages.*
- **(26-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
* *Diagram: Detailed schematic showing pin configurations and physical links.*
- **(26-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
* *Diagram: Power loop routing circuit showing active voltage paths.*
- **(26-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
* *Diagram: Configuration database schema or address map block.*
- **(26-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
* *Diagram: Electrical circuit isolating ground terminals and surges.*
- **(26-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
* *Diagram: Mounting dimension wall map indicating clearance codes.*
- **(26-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
* *Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(26-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.

** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*

- **(26-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.

** Diagram: Lab trainer schematic showing green verification check.*

Session 27: Cybersecurity

Focus: Hardening HTTP ports, disabling UPnP, HTTPS certificates.

- **(27-1) Title Slide:** Introductory splash screen for the Cybersecurity session.

** Diagram: Center-aligned splash screen with low-voltage course tag branding.*

- **(27-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.

** Diagram: Concept block diagram mapping input-process-output stages.*

- **(27-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.

** Diagram: Detailed schematic showing pin configurations and physical links.*

- **(27-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.

** Diagram: Power loop routing circuit showing active voltage paths.*

- **(27-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.

** Diagram: Configuration database schema or address map block.*

- **(27-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.

** Diagram: Electrical circuit isolating ground terminals and surges.*

- **(27-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.

** Diagram: Mounting dimension wall map indicating clearance codes.*

- **(27-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.

** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*

- **(27-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.

** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*

- **(27-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.

** Diagram: Lab trainer schematic showing green verification check.*

Session 28: Troubleshooting & Maintenance

Focus: No-video check loops, hum bars removal, IR halo cleaning.

- **(28-1) Title Slide:** Introductory splash screen for the Troubleshooting & Maintenance session.

** Diagram: Center-aligned splash screen with low-voltage course tag branding.*

- **(28-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.

** Diagram: Concept block diagram mapping input-process-output stages.*

- **(28-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(28-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(28-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(28-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(28-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(28-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(28-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(28-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Module 3 Slide Plans Outline

Session 29: Video Door Phone

Focus: 2-wire vs IP VDP setups, lock relays, SIP app routing.

- **(29-1) Title Slide:** Introductory splash screen for the Video Door Phone session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(29-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(29-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(29-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(29-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
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** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(29-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(29-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(29-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 30: Access Control

Focus: Wiegand standard green/white lines, REX buttons, EM lock NC loop.

- **(30-1) Title Slide:** Introductory splash screen for the Access Control session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(30-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.

** Diagram: Concept block diagram mapping input-process-output stages.*

- **(30-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.

** Diagram: Detailed schematic showing pin configurations and physical links.*

- **(30-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.

** Diagram: Power loop routing circuit showing active voltage paths.*

- **(30-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.

** Diagram: Configuration database schema or address map block.*

- **(30-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.

** Diagram: Electrical circuit isolating ground terminals and surges.*

- **(30-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.

** Diagram: Mounting dimension wall map indicating clearance codes.*

- **(30-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.

** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*

- **(30-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.

** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*

- **(30-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.

** Diagram: Lab trainer schematic showing green verification check.*

Session 31: Biometric Systems

Focus: Fingerprint/facial minutiae mapping, log sync on port 4370.

- **(31-1) Title Slide:** Introductory splash screen for the Biometric Systems session.

** Diagram: Center-aligned splash screen with low-voltage course tag branding.*

- **(31-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.

** Diagram: Concept block diagram mapping input-process-output stages.*

- **(31-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.

** Diagram: Detailed schematic showing pin configurations and physical links.*

- **(31-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.

** Diagram: Power loop routing circuit showing active voltage paths.*

- **(31-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.

** Diagram: Configuration database schema or address map block.*

- **(31-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.

** Diagram: Electrical circuit isolating ground terminals and surges.*

- **(31-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.

** Diagram: Mounting dimension wall map indicating clearance codes.*

- **(31-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
* *Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(31-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
* *Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(31-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
* *Diagram: Lab trainer schematic showing green verification check.*

Session 32: Smart Locks

Focus: Fail-safe vs Fail-secure, L/Z brackets, 1N4007 flyback diodes.

- **(32-1) Title Slide:** Introductory splash screen for the Smart Locks session.
* *Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(32-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
* *Diagram: Concept block diagram mapping input-process-output stages.*
- **(32-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
* *Diagram: Detailed schematic showing pin configurations and physical links.*
- **(32-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
* *Diagram: Power loop routing circuit showing active voltage paths.*
- **(32-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
* *Diagram: Configuration database schema or address map block.*
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* *Diagram: Mounting dimension wall map indicating clearance codes.*
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* *Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(32-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
* *Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(32-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
* *Diagram: Lab trainer schematic showing green verification check.*

Session 33: Boom Barrier

Focus: Ground loop vehicle detectors, balance springs, active IR photo-beams.

- **(33-1) Title Slide:** Introductory splash screen for the Boom Barrier session.

** Diagram: Center-aligned splash screen with low-voltage course tag branding.*

- **(33-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(33-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(33-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(33-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
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- **(33-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(33-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 34: Gate Automation

Focus: Sliding vs Swing motors, pinion rack gaps, remote code rolling.

- **(34-1) Title Slide:** Introductory splash screen for the Gate Automation session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(34-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(34-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(34-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
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- **(34-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
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- **(34-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(34-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 35: Intruder Alarm Systems

Focus: EOL/DEOL sensor resistor loop logic, PIR zones, siren bells.

- **(35-1) Title Slide:** Introductory splash screen for the Intruder Alarm Systems session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(35-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(35-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
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- **(35-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(35-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 36: Fire Alarm Systems

Focus: Conventional zones parallel loops, smoke photo chambers, recalls.

- **(36-1) Title Slide:** Introductory splash screen for the Fire Alarm Systems session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(36-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(36-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(36-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(36-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(36-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
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** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(36-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(36-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(36-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 37: PA BGM Systems

Focus: 100V line audio step-up distribution, paging priority overrides.

- **(37-1) Title Slide:** Introductory splash screen for the PA BGM Systems session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(37-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(37-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(37-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*

- **(37-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(37-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(37-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(37-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(37-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(37-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 38: Intercom Systems

Focus: PABX analog extensions Krone blocks, SIP IP-PBX IVR routing.

- **(38-1) Title Slide:** Introductory splash screen for the Intercom Systems session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(38-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(38-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(38-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(38-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(38-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(38-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(38-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(38-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(38-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.

** Diagram: Lab trainer schematic showing green verification check.*

Session 39: Elevator Access Control

Focus: Floor selector relay splitting matrix, cabin readers, fire loops.

- **(39-1) Title Slide:** Introductory splash screen for the Elevator Access Control session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(39-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(39-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(39-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(39-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(39-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(39-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(39-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(39-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(39-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 40: Visitor Management

Focus: OCR document scan checklists, QR passes, turnstile syncs.

- **(40-1) Title Slide:** Introductory splash screen for the Visitor Management session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(40-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(40-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*

- **(40-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(40-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
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** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(40-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(40-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 41: Hotel Lock Systems

Focus: Mifare Classic sector keys, offline RTC lock sync, room savers.

- **(41-1) Title Slide:** Introductory splash screen for the Hotel Lock Systems session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(41-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(41-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(41-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(41-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
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** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(41-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(41-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(41-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.

** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*

- **(41-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.

** Diagram: Lab trainer schematic showing green verification check.*

Session 42: Parking Management

Focus: Ticket lane dispensers loops, exit card collectors, anti-passback.

- **(42-1) Title Slide:** Introductory splash screen for the Parking Management session.

** Diagram: Center-aligned splash screen with low-voltage course tag branding.*

- **(42-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.

** Diagram: Concept block diagram mapping input-process-output stages.*

- **(42-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.

** Diagram: Detailed schematic showing pin configurations and physical links.*

- **(42-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.

** Diagram: Power loop routing circuit showing active voltage paths.*

- **(42-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.

** Diagram: Configuration database schema or address map block.*

- **(42-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.

** Diagram: Electrical circuit isolating ground terminals and surges.*

- **(42-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.

** Diagram: Mounting dimension wall map indicating clearance codes.*

- **(42-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.

** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*

- **(42-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.

** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*

- **(42-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.

** Diagram: Lab trainer schematic showing green verification check.*

Session 43: Basic BMS

Focus: DDC IO blocks terminals, PT100 temperature resistance, Modbus RS485.

- **(43-1) Title Slide:** Introductory splash screen for the Basic BMS session.

** Diagram: Center-aligned splash screen with low-voltage course tag branding.*

- **(43-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.

** Diagram: Concept block diagram mapping input-process-output stages.*

- **(43-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(43-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(43-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(43-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(43-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(43-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(43-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(43-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Module 4 Slide Plans Outline

Session 44: KNX Fundamentals

Focus: KNX green bus TP1 topology, line addressing, ETS group mappings.

- **(44-1) Title Slide:** Introductory splash screen for the KNX Fundamentals session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(44-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(44-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(44-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(44-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(44-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(44-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(44-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(44-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(44-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 45: Wi-Fi Smart Home

Focus: Neutral switches, bypass capacitors bulb loop, Tuya gateway APIs.

- **(45-1) Title Slide:** Introductory splash screen for the Wi-Fi Smart Home session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(45-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.

** Diagram: Concept block diagram mapping input-process-output stages.*

- **(45-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.

** Diagram: Detailed schematic showing pin configurations and physical links.*

- **(45-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.

** Diagram: Power loop routing circuit showing active voltage paths.*

- **(45-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.

** Diagram: Configuration database schema or address map block.*

- **(45-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.

** Diagram: Electrical circuit isolating ground terminals and surges.*

- **(45-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.

** Diagram: Mounting dimension wall map indicating clearance codes.*

- **(45-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.

** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*

- **(45-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.

** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*

- **(45-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.

** Diagram: Lab trainer schematic showing green verification check.*

Session 46: Zigbee & Z-Wave

Focus: Mesh coordinator/router/end roles, RF coexistence, self-healing.

- **(46-1) Title Slide:** Introductory splash screen for the Zigbee & Z-Wave session.

** Diagram: Center-aligned splash screen with low-voltage course tag branding.*

- **(46-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.

** Diagram: Concept block diagram mapping input-process-output stages.*

- **(46-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.

** Diagram: Detailed schematic showing pin configurations and physical links.*

- **(46-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.

** Diagram: Power loop routing circuit showing active voltage paths.*

- **(46-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.

** Diagram: Configuration database schema or address map block.*

- **(46-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.

** Diagram: Electrical circuit isolating ground terminals and surges.*

- **(46-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.

** Diagram: Mounting dimension wall map indicating clearance codes.*

- **(46-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
* *Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(46-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
* *Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(46-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
* *Diagram: Lab trainer schematic showing green verification check.*

Session 47: Matter & Thread

Focus: IPv6 Thread mesh Border Routers, Multi-Admin platform sync.

- **(47-1) Title Slide:** Introductory splash screen for the Matter & Thread session.
* *Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(47-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
* *Diagram: Concept block diagram mapping input-process-output stages.*
- **(47-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
* *Diagram: Detailed schematic showing pin configurations and physical links.*
- **(47-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
* *Diagram: Power loop routing circuit showing active voltage paths.*
- **(47-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
* *Diagram: Configuration database schema or address map block.*
- **(47-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
* *Diagram: Electrical circuit isolating ground terminals and surges.*
- **(47-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
* *Diagram: Mounting dimension wall map indicating clearance codes.*
- **(47-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
* *Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(47-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
* *Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(47-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
* *Diagram: Lab trainer schematic showing green verification check.*

Session 48: Alexa & Google Home Integration

Focus: Cloud-to-Cloud API tokens OAuth skill pairing, geofencing.

- **(48-1) Title Slide:** Introductory splash screen for the Alexa & Google Home Integration session.

** Diagram: Center-aligned splash screen with low-voltage course tag branding.*

- **(48-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(48-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(48-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(48-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(48-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(48-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(48-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(48-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(48-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 49: Home Assistant

Focus: HAOS Raspberry Pi setup, Lovelace dashboard design, YAML automations.

- **(49-1) Title Slide:** Introductory splash screen for the Home Assistant session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(49-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(49-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(49-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(49-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(49-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*

- **(49-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(49-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
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- **(49-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(49-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 50: Smart Lighting & Curtain

Focus: Leading/trailing phase cut AC dimmers, curtain torque calibrations.

- **(50-1) Title Slide:** Introductory splash screen for the Smart Lighting & Curtain session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(50-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(50-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(50-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(50-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(50-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(50-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(50-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(50-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(50-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 51: Smart Irrigation & Energy Monitoring

Focus: 24VAC water solenoids, CT clamp calibration, HA energy panels.

- **(51-1) Title Slide:** Introductory splash screen for the Smart Irrigation & Energy Monitoring session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(51-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(51-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(51-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(51-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
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- **(51-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
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** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(51-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(51-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 52: Solar CCTV & IoT Fundamentals

Focus: Solar panels calculations, MPPT charging, 4G SIM APN routing.

- **(52-1) Title Slide:** Introductory splash screen for the Solar CCTV & IoT Fundamentals session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(52-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(52-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(52-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*

- **(52-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
* *Diagram: Configuration database schema or address map block.*
- **(52-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
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* *Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(52-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
* *Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(52-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
* *Diagram: Lab trainer schematic showing green verification check.*

Session 53: Site Survey & BOQ/Quotation

Focus: Site blue-print survey checklist, Excel BOQs, margin pricing.

- **(53-1) Title Slide:** Introductory splash screen for the Site Survey & BOQ/Quotation session.
* *Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(53-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
* *Diagram: Concept block diagram mapping input-process-output stages.*
- **(53-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
* *Diagram: Detailed schematic showing pin configurations and physical links.*
- **(53-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
* *Diagram: Power loop routing circuit showing active voltage paths.*
- **(53-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
* *Diagram: Configuration database schema or address map block.*
- **(53-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
* *Diagram: Electrical circuit isolating ground terminals and surges.*
- **(53-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
* *Diagram: Mounting dimension wall map indicating clearance codes.*
- **(53-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
* *Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(53-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
* *Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(53-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.

** Diagram: Lab trainer schematic showing green verification check.*

Session 54: Project Planning & Installation Standards

Focus: 40% conduit fills, cable labeling tags, switch wall heights.

- **(54-1) Title Slide:** Introductory splash screen for the Project Planning & Installation Standards session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(54-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(54-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(54-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(54-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
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- **(54-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
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** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(54-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(54-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 55: Testing, Commissioning & AMC

Focus: Continuity insulation megger logs, handovers, SLA AMC tiers.

- **(55-1) Title Slide:** Introductory splash screen for the Testing, Commissioning & AMC session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(55-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(55-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*

- **(55-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
* *Diagram: Power loop routing circuit showing active voltage paths.*
- **(55-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
* *Diagram: Configuration database schema or address map block.*
- **(55-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
* *Diagram: Electrical circuit isolating ground terminals and surges.*
- **(55-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
* *Diagram: Mounting dimension wall map indicating clearance codes.*
- **(55-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
* *Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(55-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
* *Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(55-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
* *Diagram: Lab trainer schematic showing green verification check.*

Session 56: Customer Support & Sales/Negotiation

Focus: Service priority SLAs (P1-P3), price objections, scope changes.

- **(56-1) Title Slide:** Introductory splash screen for the Customer Support & Sales/Negotiation session.
* *Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(56-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
* *Diagram: Concept block diagram mapping input-process-output stages.*
- **(56-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
* *Diagram: Detailed schematic showing pin configurations and physical links.*
- **(56-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
* *Diagram: Power loop routing circuit showing active voltage paths.*
- **(56-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
* *Diagram: Configuration database schema or address map block.*
- **(56-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
* *Diagram: Electrical circuit isolating ground terminals and surges.*
- **(56-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
* *Diagram: Mounting dimension wall map indicating clearance codes.*
- **(56-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
* *Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(56-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.

** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*

- **(56-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.

** Diagram: Lab trainer schematic showing green verification check.*

Session 57: Digital Marketing & Business Automation

Focus: Google Business local SEO, WhatsApp catalog auto-responses.

- **(57-1) Title Slide:** Introductory splash screen for the Digital Marketing & Business Automation session.

** Diagram: Center-aligned splash screen with low-voltage course tag branding.*

- **(57-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.

** Diagram: Concept block diagram mapping input-process-output stages.*

- **(57-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.

** Diagram: Detailed schematic showing pin configurations and physical links.*

- **(57-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.

** Diagram: Power loop routing circuit showing active voltage paths.*

- **(57-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.

** Diagram: Configuration database schema or address map block.*

- **(57-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.

** Diagram: Electrical circuit isolating ground terminals and surges.*

- **(57-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.

** Diagram: Mounting dimension wall map indicating clearance codes.*

- **(57-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.

** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*

- **(57-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.

** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*

- **(57-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.

** Diagram: Lab trainer schematic showing green verification check.*

Session 58: Business Setup & Entrepreneurship

Focus: Company LLC structure trade license, distributor accounts credit.

- **(58-1) Title Slide:** Introductory splash screen for the Business Setup & Entrepreneurship session.

** Diagram: Center-aligned splash screen with low-voltage course tag branding.*

- **(58-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.

** Diagram: Concept block diagram mapping input-process-output stages.*

- **(58-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(58-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(58-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(58-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(58-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(58-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.
** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*
- **(58-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.
** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*
- **(58-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.
** Diagram: Lab trainer schematic showing green verification check.*

Session 59: Placement Training & Career Prep

Focus: Technical project resumes, LinkedIn search keywords, interviews.

- **(59-1) Title Slide:** Introductory splash screen for the Placement Training & Career Prep session.
** Diagram: Center-aligned splash screen with low-voltage course tag branding.*
- **(59-2) Core Operational Concepts:** Theoretical overview of core rules and mechanics.
** Diagram: Concept block diagram mapping input-process-output stages.*
- **(59-3) Hardware & Wiring Schematics:** Cabling paths, terminal blocks, and port pinouts.
** Diagram: Detailed schematic showing pin configurations and physical links.*
- **(59-4) Electrical Power Sizing:** Sizing power margins, currents, voltages, and backups.
** Diagram: Power loop routing circuit showing active voltage paths.*
- **(59-5) Software Configuration & Setting:** Setting IP addresses, register configurations, and logical rules.
** Diagram: Configuration database schema or address map block.*
- **(59-6) Protections & Surges:** Isolating noise, grounding shield lines, and surge suppressors.
** Diagram: Electrical circuit isolating ground terminals and surges.*
- **(59-7) Installation Standards & Codes:** Height regulations, conduit fills, and labeling codes.
** Diagram: Mounting dimension wall map indicating clearance codes.*
- **(59-8) Diagnostics & Troubleshooting:** Tracing failures, resolving crashes, and continuity logs.

** Diagram: Troubleshooting workflow checklist mapping issues to solutions.*

- **(59-9) Safety Interlocks & Overrides:** Master override relays, mechanical manual releases.

** Diagram: Interlock loop wiring showing fire panel emergency disconnects.*

- **(59-10) Lab Work checklist:** Practical exercise testing, sign-off parameters.

** Diagram: Lab trainer schematic showing green verification check.*